

MEDIAN OF PEER MULTIPLES

The Nasdaq data, and your first skill

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Thu Aug 27, 2026

TODAY

Three Things

- 1 The data — twenty-five years of fundamentals and prices, and how to reach it
- 2 Skills — how you stop re-explaining yourself to the machine
- 3 Build one — a valuation skill that prices a company off its peers

By the end you will have something in your group's repository that any of you can point at a ticker.

The Oldest Valuation Method There Is

A company is worth roughly what companies like it are worth.

No forecast, no discount rate, no terminal value. Find the peers, take the middle of their multiples, apply it. It is what a banker does on the back of an envelope, and it is a real benchmark — hard to beat and easy to check.

The whole method is three decisions: which multiple, which peers, and what to do with the ones that do not make sense.

THE DATA

The Rice Business Data Portal

What is in it

Every NYSE and Nasdaq listed common stock, beginning in 2000.

Where it comes from

Licensed by Rice Business from Nasdaq Data Link.

What it is

A DuckDB database. Queries are read-only — `SELECT` and nothing else.

data-portal.rice-business.org

- 1 Register at the portal with your `rice.edu` address
- 2 You are issued an access token, good for six months
- 3 Put it where the lab can find it — not in a file you commit

The token is yours. A token in a public repository is a token somebody else is using by Thursday.

Get Your Token

Three Doors Into It

The web interface

A query window with an AI assistant that writes the SQL for you. Good for looking around.

The Python client

A library for notebooks and scripts. This is what your analysis will use.

The MCP server

GLM queries the database directly, without you writing anything.

The third one is how we will work today. It is also the one that makes it easy to stop paying attention — so we will keep looking at the SQL.

What Is in There

TABLE	CONTENTS
sep	Daily open, high, low, close, adjusted close, volume
daily	Daily valuation ratios and market cap
metrics	Daily moving averages and other statistics
sf1	10-K and 10-Q data, and pre-computed financial ratios
sf2	Insider trades, from Form 4
tickers	Industry, sector, exchange, size category
actions	Mergers, splits, dividends, ticker changes
events	SEC 8-K filings and material corporate events

Today We Need Three of Them

daily

pe , pb , ps , ev ,
evebit , evebitda ,
marketcap — one
row per stock per
day.

tickers

sector , industry ,
exchange ,
scalemarketcap .
This is what makes a
peer group a peer
group.

sf1

The filings. Revenue,
earnings, book
value, and the ratios
computed from
them.

	ANNUAL	QUARTERLY	TRAILING FOUR QUARTERS
As most recently restated	MRY	MRQ	MRT
As originally reported	ARY	ARQ	ART

Today we use **MRY** — annual, most recent restatement. It is the cleanest view of what a company earned. When we start backtesting, we will have to switch, because **MRY** tells you today what nobody knew at the time.

sf1 Has Dimensions

Three Things That Trip Everyone

Dates are text

Cast before you compare or sort:

```
date::DATE, reportperiod::DATE
```

`sf1` has no `date`

Use `reportperiod` for the fiscal period end, `datekey` for the filing date

Market cap is in thousands

In `daily`. A \$2T company shows up as 2,000,000,000

Each of these produces an answer, not an error. That is what makes them expensive.

PULLING IT

Start by Looking Around

TELL GLM

Using the data portal, show me the columns in `daily` and `tickers`, and give me five rows of each.

Before any analysis, look at the actual rows. Half of the mistakes in this course get caught here, in thirty seconds, by someone who looked.

TELL GLM

For the most recent date in `daily`, get `ticker`, `pe`, and `marketcap` for every stock. Join `sector` and `industry` from `tickers`. Then pull the `MRY` rows from `sf1` for the most recent fiscal year and bring in revenue, net income, and shares outstanding. Show me the SQL before you run it.

That last sentence is the entire lesson from Tuesday, in six words

The Pull

Read the SQL

You are not checking whether it is elegant. You are checking four things.

Did it filter the dimension?

Without `dimension = 'MRY'` you get six rows per company-year

Did it cast the dates?

Text comparison on dates goes wrong quietly

Is the join key right?

`ticker` is not unique across time — see `actions`

How many rows came back?

You know roughly how many listed stocks there are

What a P/E of -12 Means

Nothing. It means the company lost money.

A negative P/E is not a cheap stock. It is a stock with no E. Left in the sample it drags the median in the wrong direction and makes the cheapest-looking companies the ones that are losing the most.

TELL GLM

Drop the negative P/E rows, and show me how many you dropped by sector.

Look Before You Compute

TELL GLM

Histogram of P/E across all stocks. Then the same by sector, and a table of the count, median, and the 10th and 90th percentiles in each.

You are looking for two things: how fat the right tail is, and whether the sectors actually differ. If they do not, the peer group is doing no work and the method has nothing to offer.

MEDIAN OF PEERS

Why the Median

THE MEAN

One company with a P/E of 900 — a real company that barely broke even — and the average for the industry is meaningless.

THE MEDIAN

Unmoved by the tail. With ratios, where the numerator is bounded below and the denominator can approach zero, this is not a refinement. It is the difference between a number and noise.

Sector

Eleven of them.
Always enough
companies, often
not comparable
ones.

Industry

Much finer.
Sometimes four
companies, and then
the median is one
company's P/E.

Industry and size

Better in principle.
Cuts the group
again, so you need
to watch the count.

There is no correct answer. There is a choice you make, write down,
and defend when you present.

Which Peers?

The Calculation

- 1 Group the stocks by peer set, and drop the ones with no meaningful P/E
- 2 Take the median P/E in each group
- 3 Multiply it by the company's own earnings per share
- 4 Compare with the price it actually trades at

The gap is the whole output. A company trading well below its peer median is either cheap or different, and the method cannot tell you which.

Now Do It

TELL GLM

Compute the median P/E within each industry. For every stock, report its own P/E, its industry median, and the ratio of the two. Show me the twenty most expensive and the twenty cheapest by that ratio.

Read the two lists. If the cheap list is all one industry, or all tiny companies, something in the peer definition is doing work you did not intend.

SKILLS

Stop Re-Explaining Yourself

Always on

Applies to every conversation you have.

Who you are, how you want to be written to, what never to do.

Per project

Applies inside one folder.

This course, this dataset, its quirks.

When relevant

A skill: read only when what you asked matches what it is for.

Three scopes, and the third is the interesting one — because it costs nothing until it applies.

A Skill Is Written Instructions, Filed Where They Can Be Found

A folder with a `SKILL.md` in it. Two parts: a line saying when to use it, and the instructions for how.

At startup

It reads only the name and the one-line description — a few dozen words per skill.

When it matches

Only then does it open the file and read the rest.

Which is why you can have many

A hundred installed skills cost nothing until one of them is relevant.

Not a plugin, and not code. A written procedure.

CLAUDE.md Versus a Skill

CLAUDE.md — ALWAYS READ

What is always true here:
what the project is, where
things live, the
conventions, what never to
do.

Read at the start of every
session in that folder.

A SKILL — READ WHEN RELEVANT

A procedure for a
particular kind of job.

Standing facts go in
CLAUDE.md. Procedures
go in skills.

Ask GLM to write the first CLAUDE.md for you — it will read the folder
and draft it.

The skill is a file. The file is in the repository.

Which means the decision is made once, by one of you, and the other two get it by pulling. It also means the decision is visible: when you present, the skill is the record of what you chose and why.

Every ad-hoc instruction typed into a prompt window dies with that conversation.

Why This Matters to a Group

BUILD ONE

Get It Right Once, Then Save It

- 1 Get the peer-median valuation working the way you want it, in a conversation
- 2 Say: save that as a skill
- 3 Start a fresh session and ask it to value a different company

Step 3 is the test. If it needs to be told anything you already decided, the skill is not finished.

What Has to Be Written Down

The decisions

Which multiple.
Which peer
definition. Median,
not mean. What the
minimum group size
is.

The traps

`dimension = 'MRY'`.
Cast the dates.
Market cap is in
thousands. Drop
negative earnings.

The output

What it should hand
back every time: the
peer set and its size,
the median, the
implied value, the
gap.

The middle card is the one that pays for itself. Those four traps will otherwise bite each of you, separately, for six weeks.

Exercise

- 1 Build the skill in your group's repository
- 2 Test it on a company none of you used while building it
- 3 Then test it on a bank, and on a company that lost money last year
- 4 Commit and push it

Steps 2 and 3 are where you find out what you left implicit. Write down what broke — it is what you will present on Tuesday.

The industry of one

A peer group with three companies in it has a median that is one company's opinion.

Financials

P/E means something different for a bank. So does book value.

Growth

Two companies in the same industry with different growth rates should not trade at the same multiple. The method assumes they do.

The third one is what Thursday of next week is about: predicting the

Where It Will Fail

BEFORE TUESDAY

Three Things

- 1 Finish the skill, and push it
- 2 Run it on five companies you know something about, and see whether the answers are sane
- 3 Bring the one case where it was most obviously wrong

Tuesday we turn this into a trade: rank every stock by the gap, buy the cheap ones, sell the expensive ones, and find out what that would have earned.